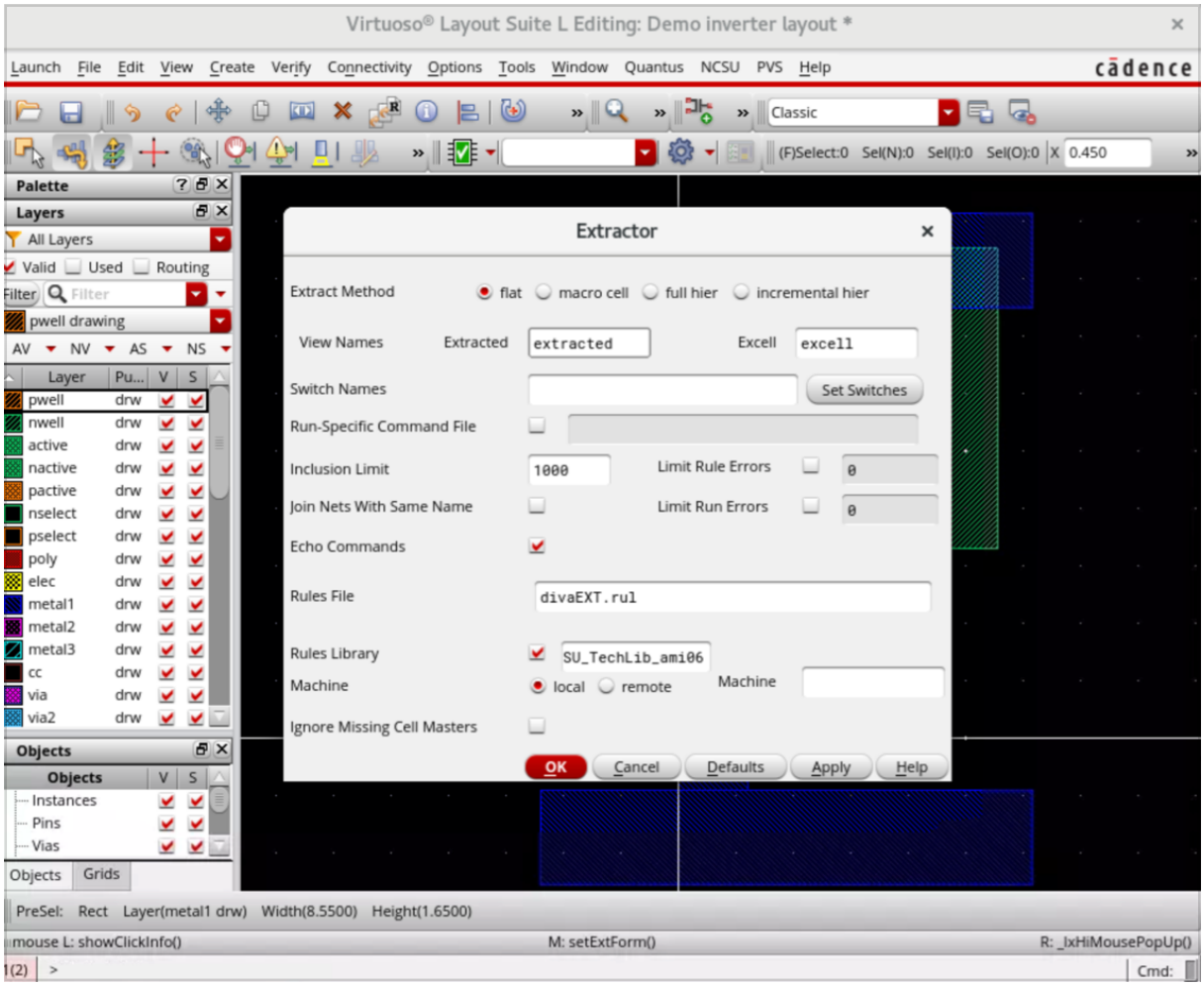
Part 3: Layout Versus Schematic(LVS) Verification For Inverter

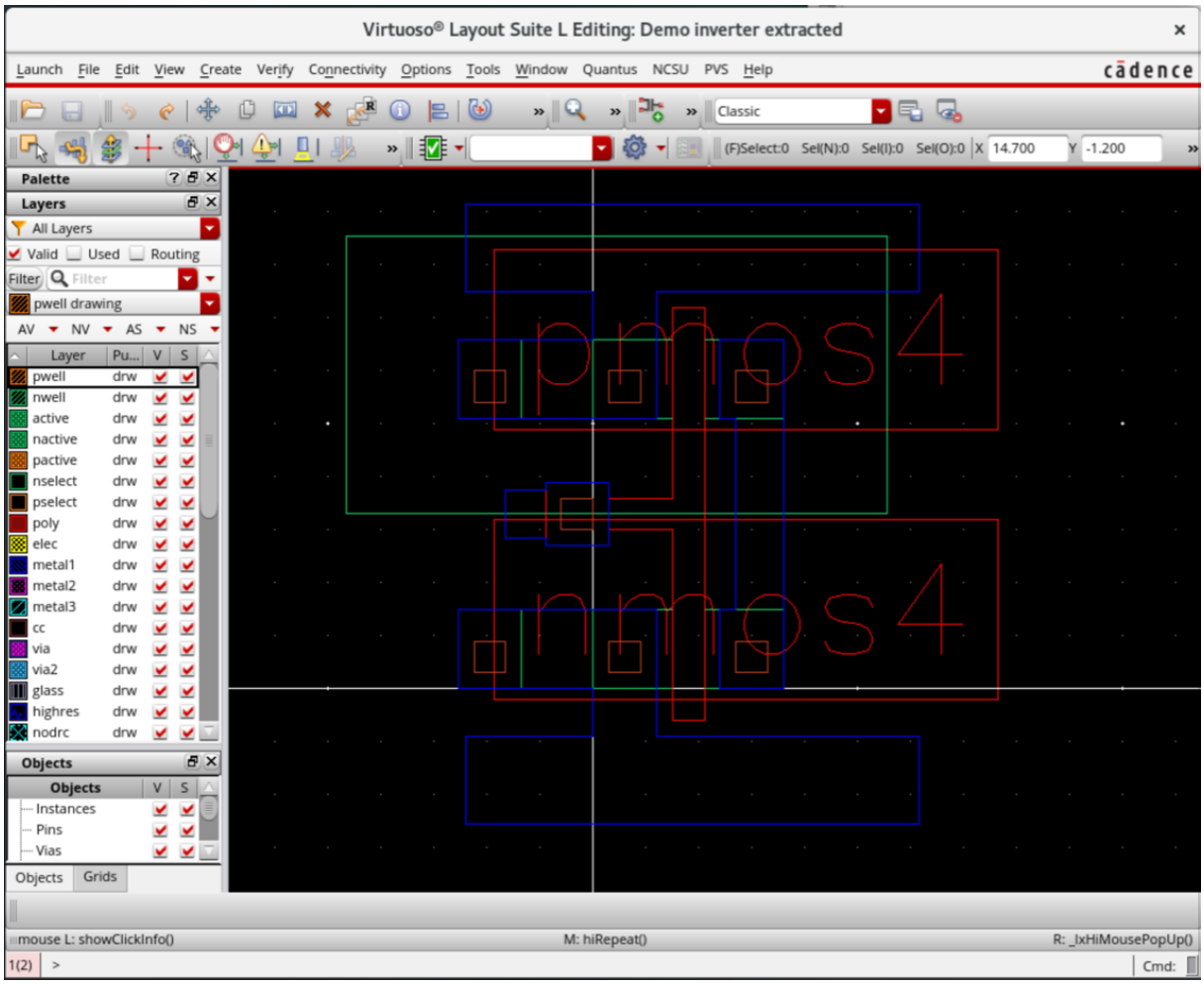
We know that the schematic and layout is just two different forms of describing the circuit. But here is a question: how can we know the layout just represent the schematic? The DRC ensures that the layout passes through the rules designed for faultless fabrication. However, it does not guarantee if it really represents the circuit you desire to fabricate. So, we need to compare the layout and schematic to make sure they represent the same circuit, which means the layout and schematic have the same parameter of transistors and other elements. This is what we are going to do during today' s lab.

1. First we need to extract the connectivities and parasitic capacitances in our layout design. So, in the directory of inverter, you open the cadence. Then you open the layout of the inverter, choose Extract... under the Verify menu.



2. Your layout will be extracted and you can check the CIW to ensure the extraction is successful.

3. a. Open the extracted view of the cell in edit mode from the Library Manager window. The extracted view will look something like this:

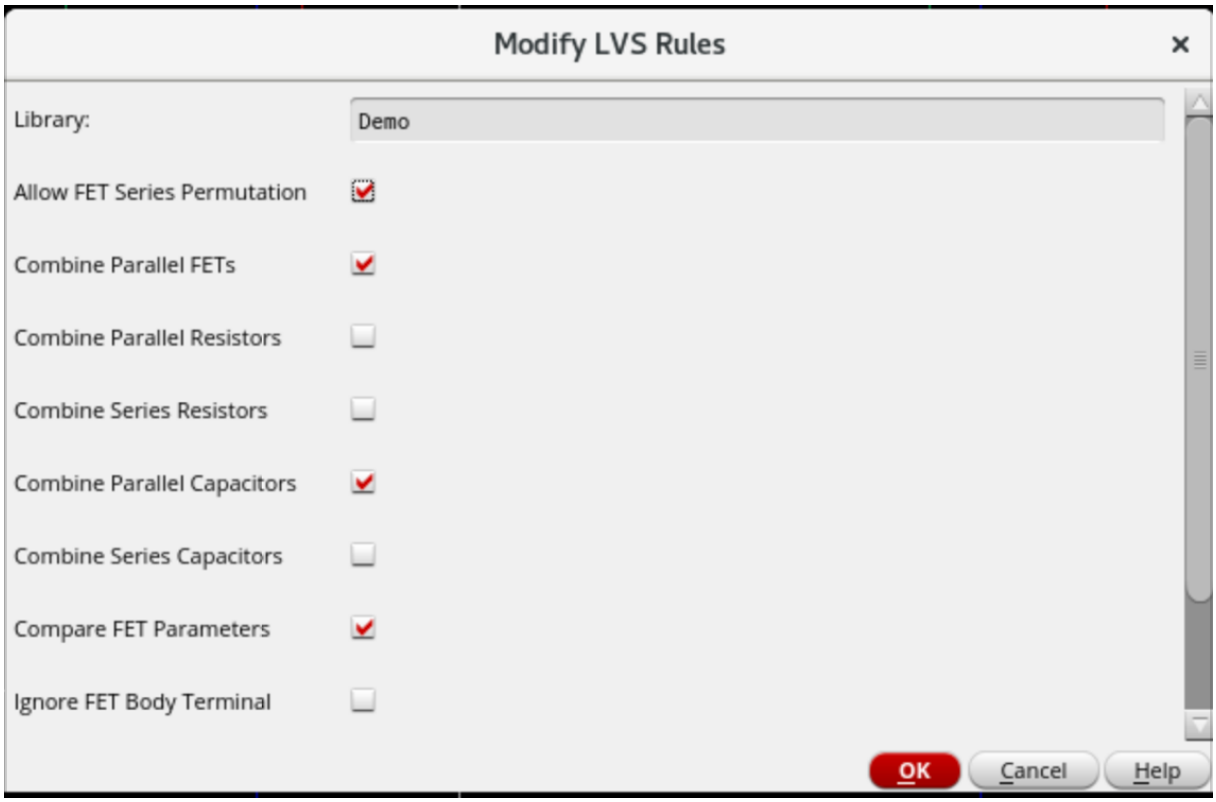


You will see either “nmos4” or “pmos4” written on it. This means that the cadence can recognize the layout of transistor. You can press Shift-F to see the symbols for the active and passive devices appear in the extraction view window.

So the next step is going to do the LVS to check whether the width and length of transistor are same for layout and schematic. You may get some errors if the size doesn’t match.

b. Before doing the LVS check, we need to choose the options to check. click on NCSU ➔ Modify LVS Rules … select the four options:

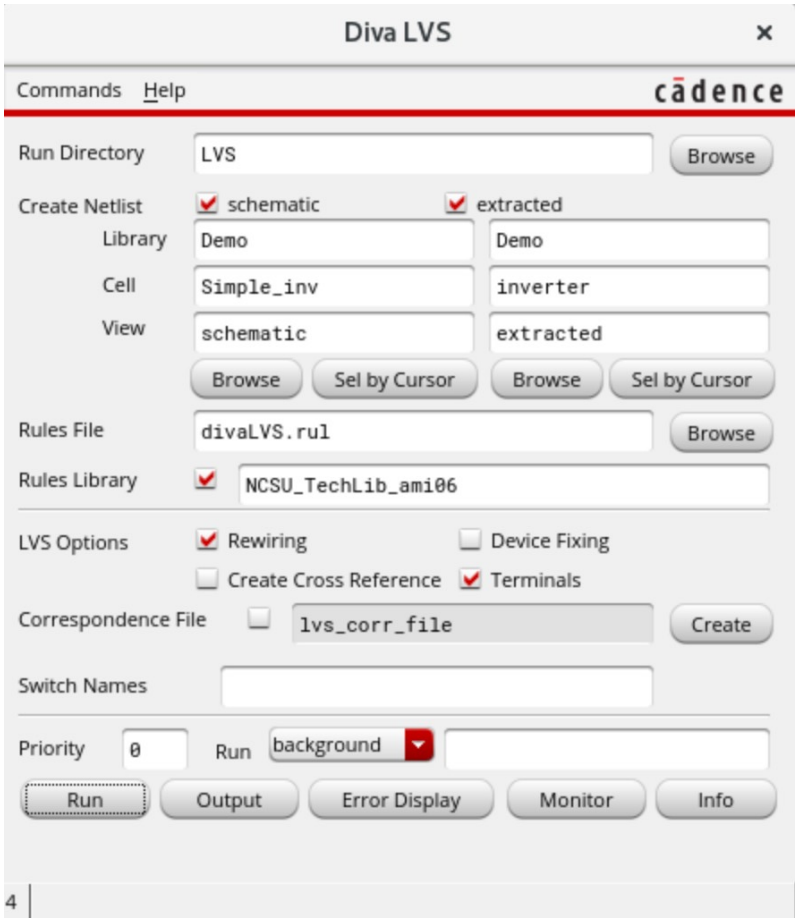
Allow FET Series Permutations,
Combine Parallel FETs,
Combine Parallel Capacitors,
Compare FET Parameters.



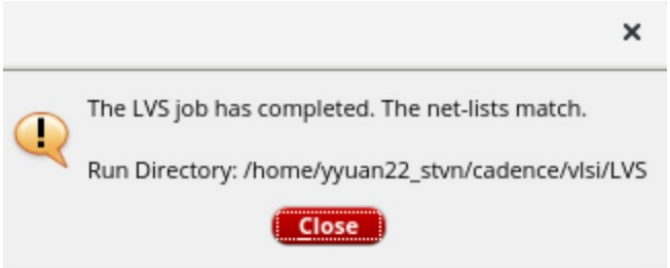
c. In extracted window, we can choose LVS... under the Verify menu, and input the cell name and the directory,

In schematic part, select the inverter schematic you created in Lab1

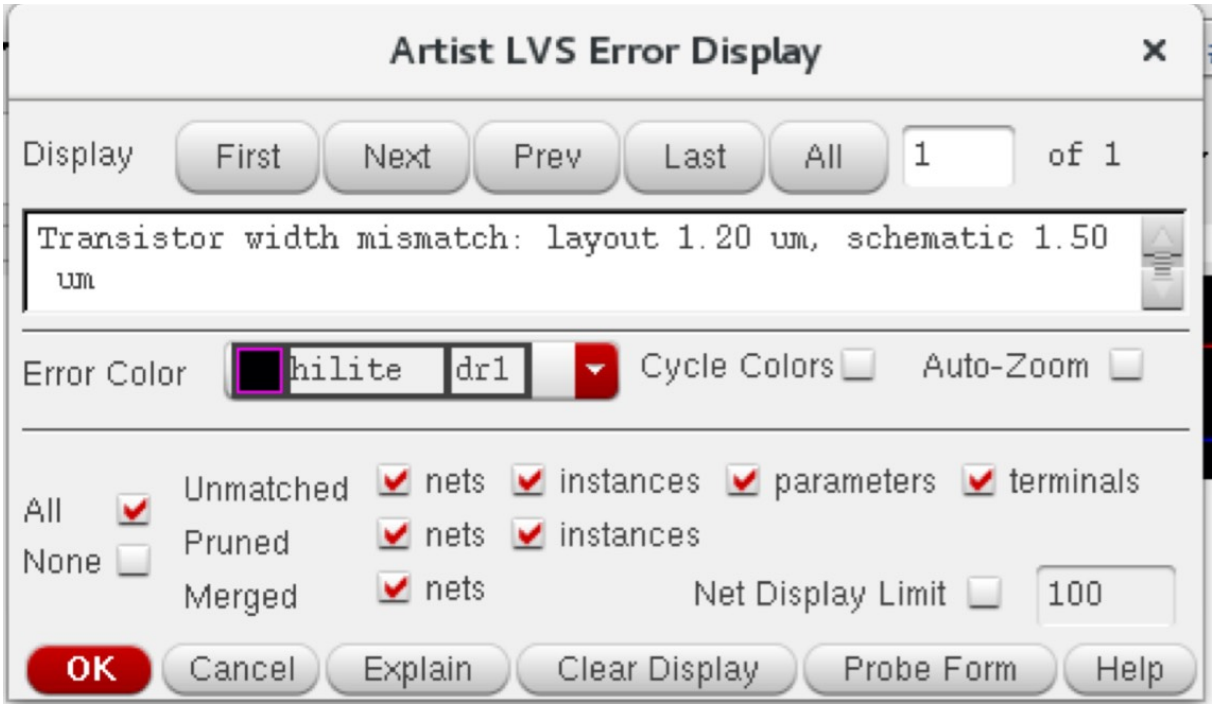
click OK. Click on the Run button and wait.



4. A window will appear to tell you whether the LVS is successful or not.



5. You can click on Error Display in the LVS menu to see the errors.



7. So we need to modify the layout or schematic. Here we choose to modify the layout of inverter. Since the error shows that the width mismatch, we need to increase the width of active layer for both pmos and nmos to 1.5um. Remember that each time you modify the layout, DRC check is required to make sure the design for layout is correct. Then we can extract the new layout and do LVS check again. By repeating this process until both DRC check for layout and LVS check for extract are correct, we can say the layout design is perfectly matched to the schematic view.